INDIAN PREMIER LEAGUE · SEASONS 2008–2024

IPL Data Analytics: From Raw Data to Player Statistics

Cleaning, dashboard and recommendations across 17 seasons

SUBJECT NAME
Data Analytics & Visualization

SUBJECT CODE
1IT508

PRESENTED BY
Team Nexus | B.Tech IT, R. N. G. Patel Institute of Technology, Bardoli

PEN NUMBERS
240840116025 · 240840116002 · 240840116062 · 240840116051 · 240840116118

Dataset:deliveries.csv

INDIAN PREMIER LEAGUE

R.N.G. PATEL INSTITUTE OF TECHNOLOGY
BARDOLI-2010
RNGPIT

INSIGHTVISTA

Problem & Dataset Observation

THE PROBLEM

Data Inconsistency Across 17 Seasons

IPL data spans 17 seasons delivered as separate, unstandardized files. Without extensive cleaning, season-by-season comparisons of teams, players, and venues are highly unreliable.

17 Seasons of Historical Data

Requires unified structure for analytical modeling

Critical Requirement

Rigorous data transformation needed before visualization and player stats evaluation.

KEY OBSERVATIONS

01 **Table Structure** — Dataset split into ball-by-ball deliveries.

02 **Team Name Evolution** — Franchise rebrandings over time require standardization (e.g. Delhi Capitals).

03 **Missing Values** — Nulls and blanks present in key fields including fielders.

04 **Inconsistent Formats** — Variations in date structures across datasets.

05 **Total Rows:2,60,920**
Total Columns:17

Raw Data to Proper Data: Steps

01

Load Data

Load the raw data into Power BI software

02

Handle Missing Values

Replace empty space/NA from many columns to None

03

Standardise Teams

Unify renamed teams, e.g. Delhi Daredevils to Delhi Capitals

04

Derive Fields

Added new columns such as season, measures: fours, sixes, total runs and wickets

05

Remove Duplicates

Correct data types and clean duplicate records

06

Validate & Visualize

Validate dataset and start visualization phase for analysis

Final Clean Dataset

Field	Description
Season	IPL year, 2008 to 2024
Batting team/Bowling team	Standardised team names
Innings	First or Second innings
Overs	Number of over going on
Batting runs	Runs scored by batting team
Total runs	Run per ball
Batter / Bowler / Runs	Ball-by-ball delivery data

Seasons

17

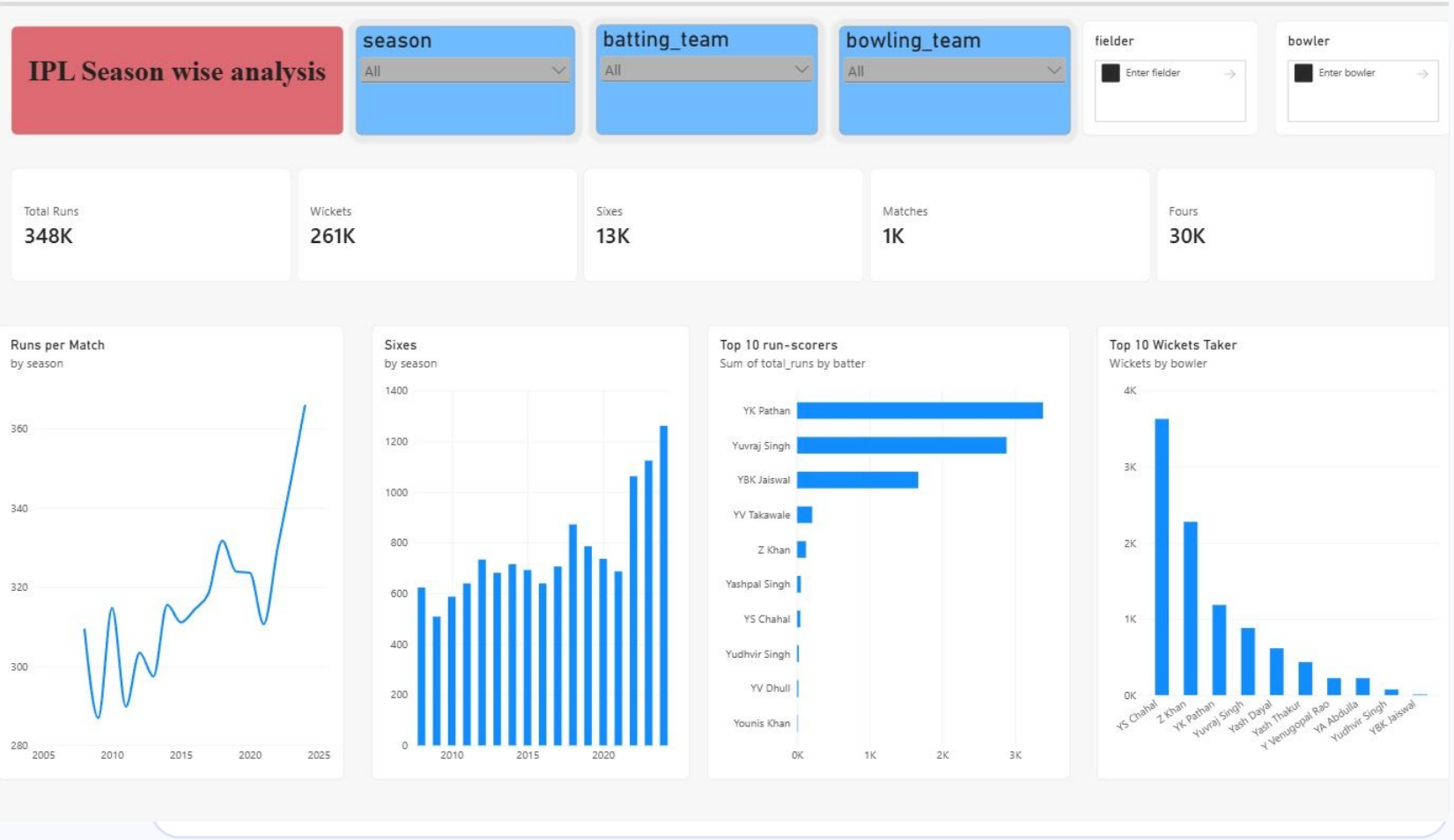
Matches

[1,095]

Deliveries

[2,60,920]

Dashboard & Insights



RUNS

348K+ Total Runs

Overall runs accumulated across 17 IPL seasons.

SIXES

13K+ Boundary Sixes

Consistent rise in boundary hitting and strike rates.

WICKETS

261K+ Wickets Taken

Ball-by-ball delivery analytics tracking bowler performance.

IPL Season wise analysis

season
All

batting_team
All

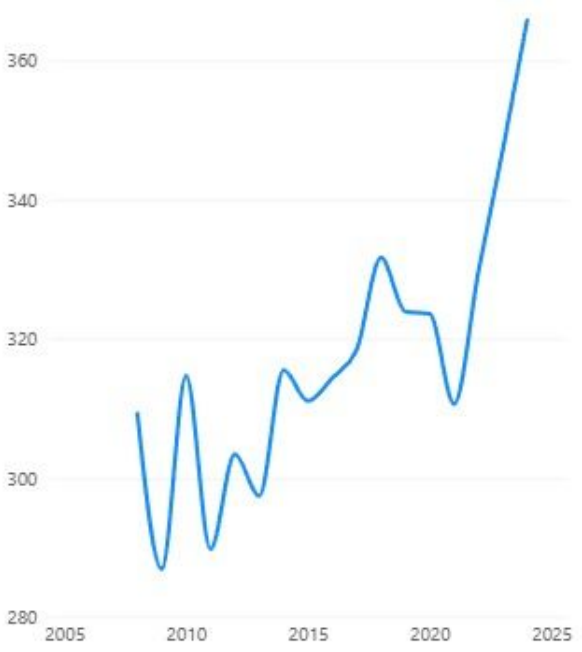
bowling_team
All

fielder
Enter fielder

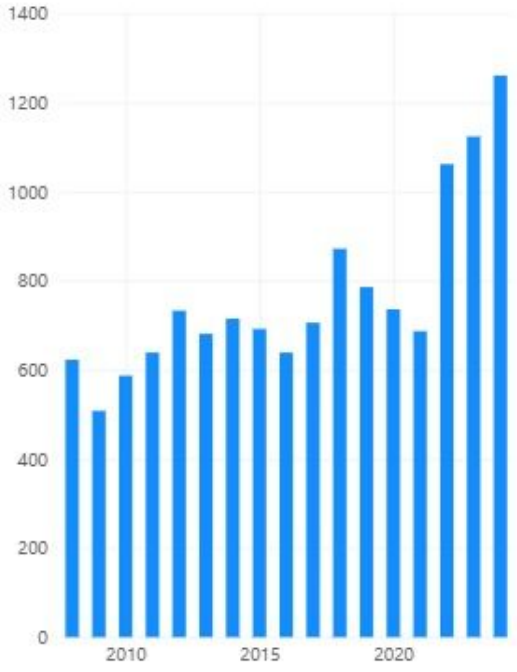
bowler
Enter bowler

Total Runs	Wickets	Sixes	Matches	Fours
348K	261K	13K	1K	30K

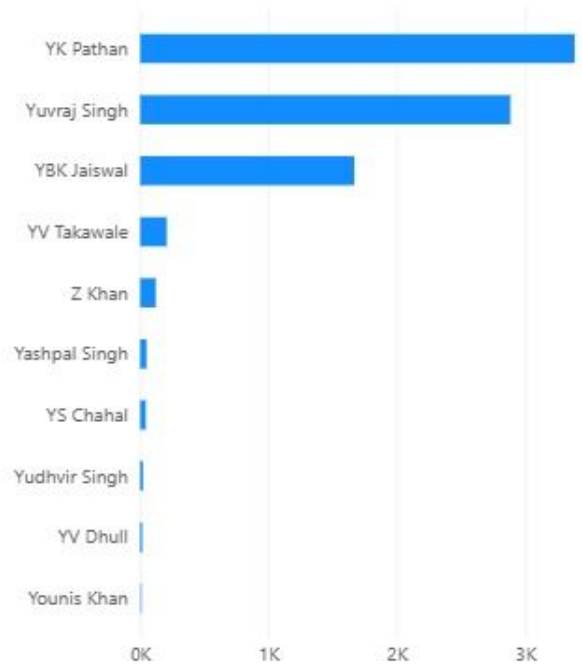
Runs per Match
by season



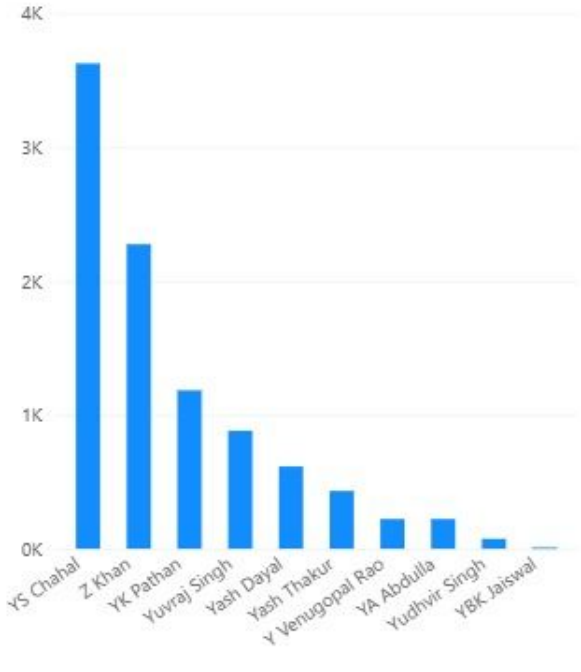
Sixes
by season



Top 10 run-scorers
Sum of total_runs by batter



Top 10 Wickets Taker
Wickets by bowler



Business Recommendations

TEAM & AUCTION STRATEGY

1. Power-Hitting Core: Prioritize high SR batters. Sixes rose 508 → 1,261/yr; runs/match 287 → 366. Gayle's 359 sixes show impact.

2. Death-Over Bowling: Overs 17–19 peak in scoring. Target low-economy death & yorker specialists.

3. Proven Consistency: Value reliable multi-season performers (Kohli: 8,014 runs) alongside power hitters.

4. Wicket-Taking Attack: Prioritize wicket-takers (Chahal: 205 wkts) to halt middle-over scoring.

5. Strategic Chasing: Chasing teams win ~53% (571 vs 507 wins); default to chase at toss.

BROADCASTERS & LEAGUE

6. Market High-Scoring Product: Promote sixes boom & big chases. Feature sponsor segments like "Max of the Match".

7. Prime Time Ad Placement: Position high-value ad slots and highlight reels around overs 17–20 (highest scoring & tension phase).

8. Protect Competitive Balance: Monitor bowler impact & pitch standards to ensure balanced bat vs ball contests.

ANALYTICS & NEXT STEPS

9. Enrich Data Integration: Join matches.csv to evaluate venue dynamics, toss impact, and venue-specific win %.

10. Key Dashboard KPIs: Track runs/match, sixes/match, death RPO, chasing win %, and top-performer run share.

Critical Caveats:

- Correlation ≠ causation (e.g., Impact Player rule influence).
- 53% chasing edge is a modest edge, not a rule.
- Financial/auction strategies require salary data.

THANK YOU